Spec-Driven Development

Arquitectura de Software II

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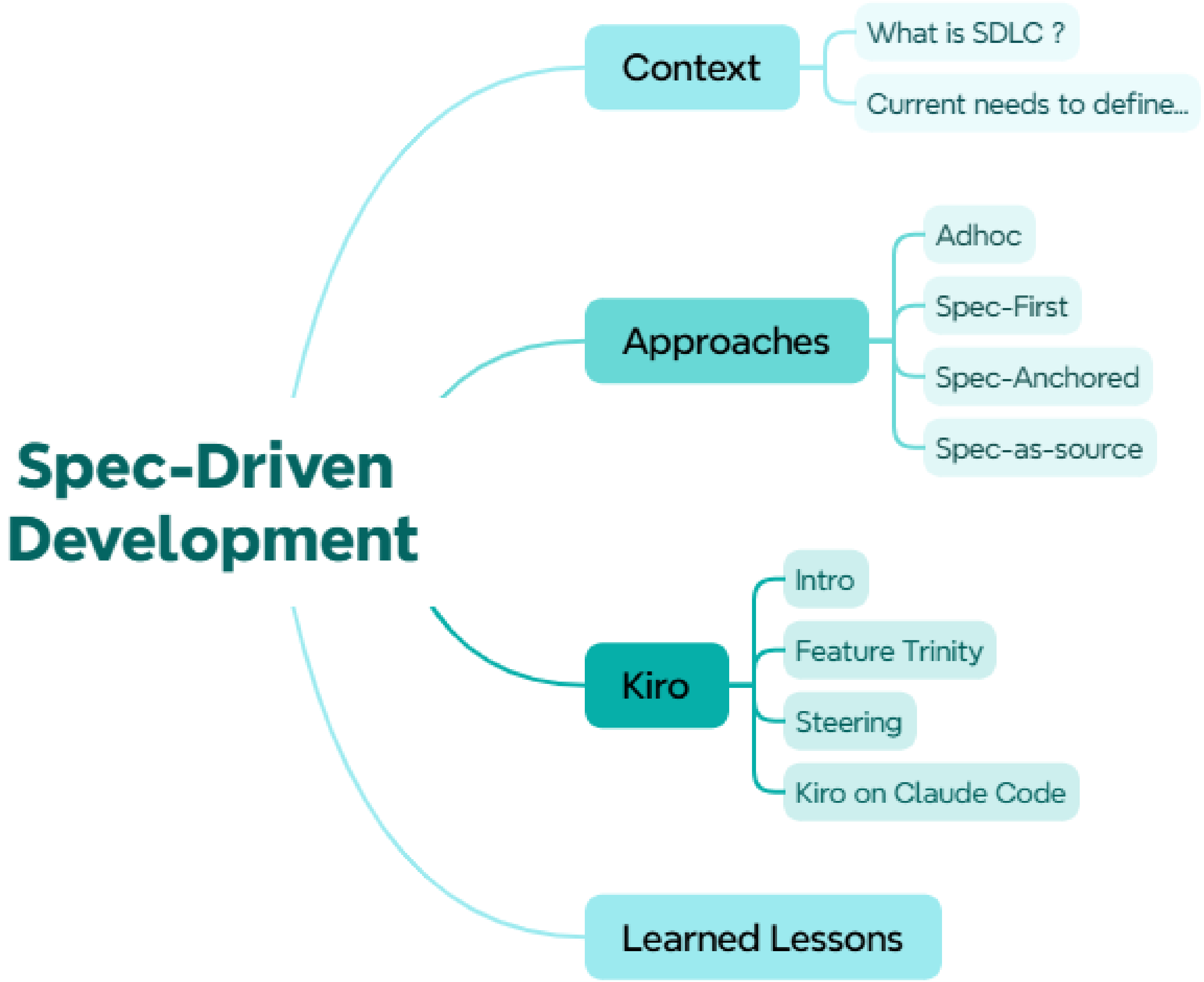
Agenda



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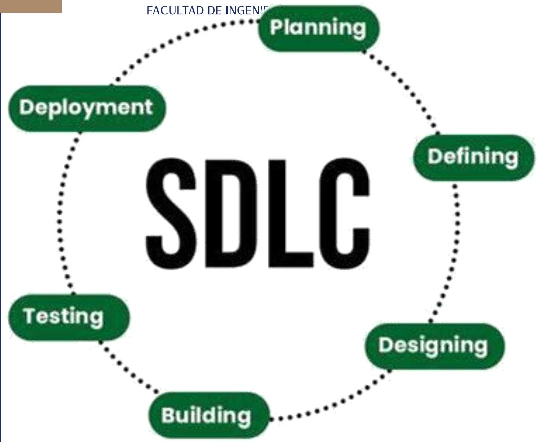
What is SDLC ?



Vibe Coding

Reactive, prompt-heavy and fragile

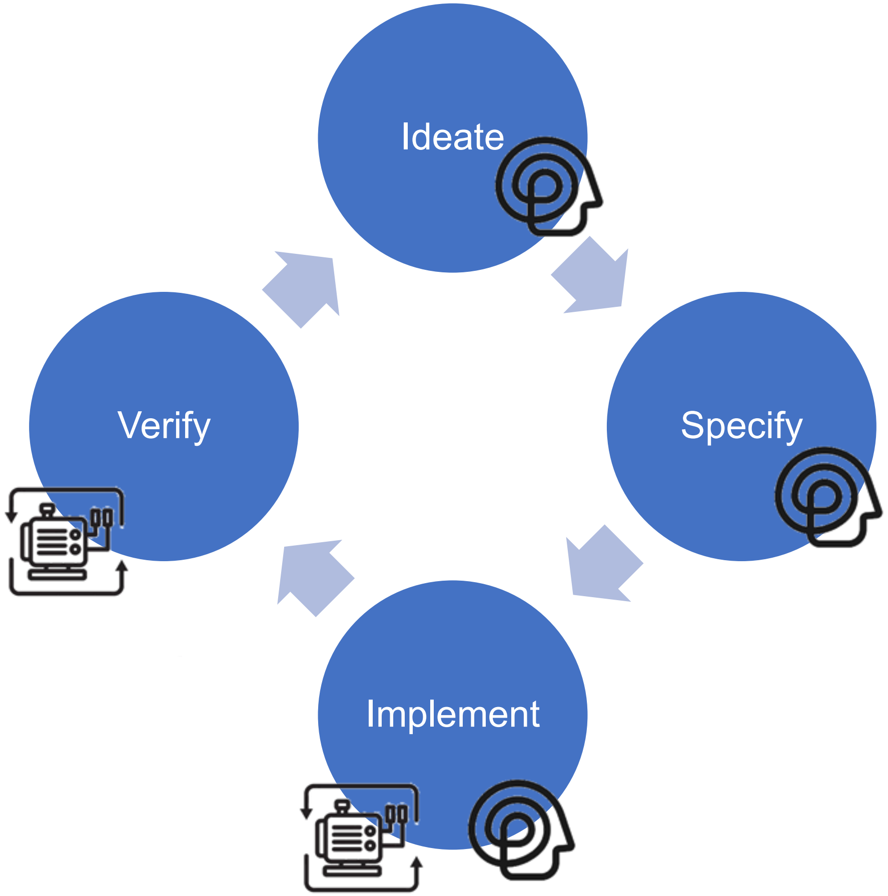
SDD

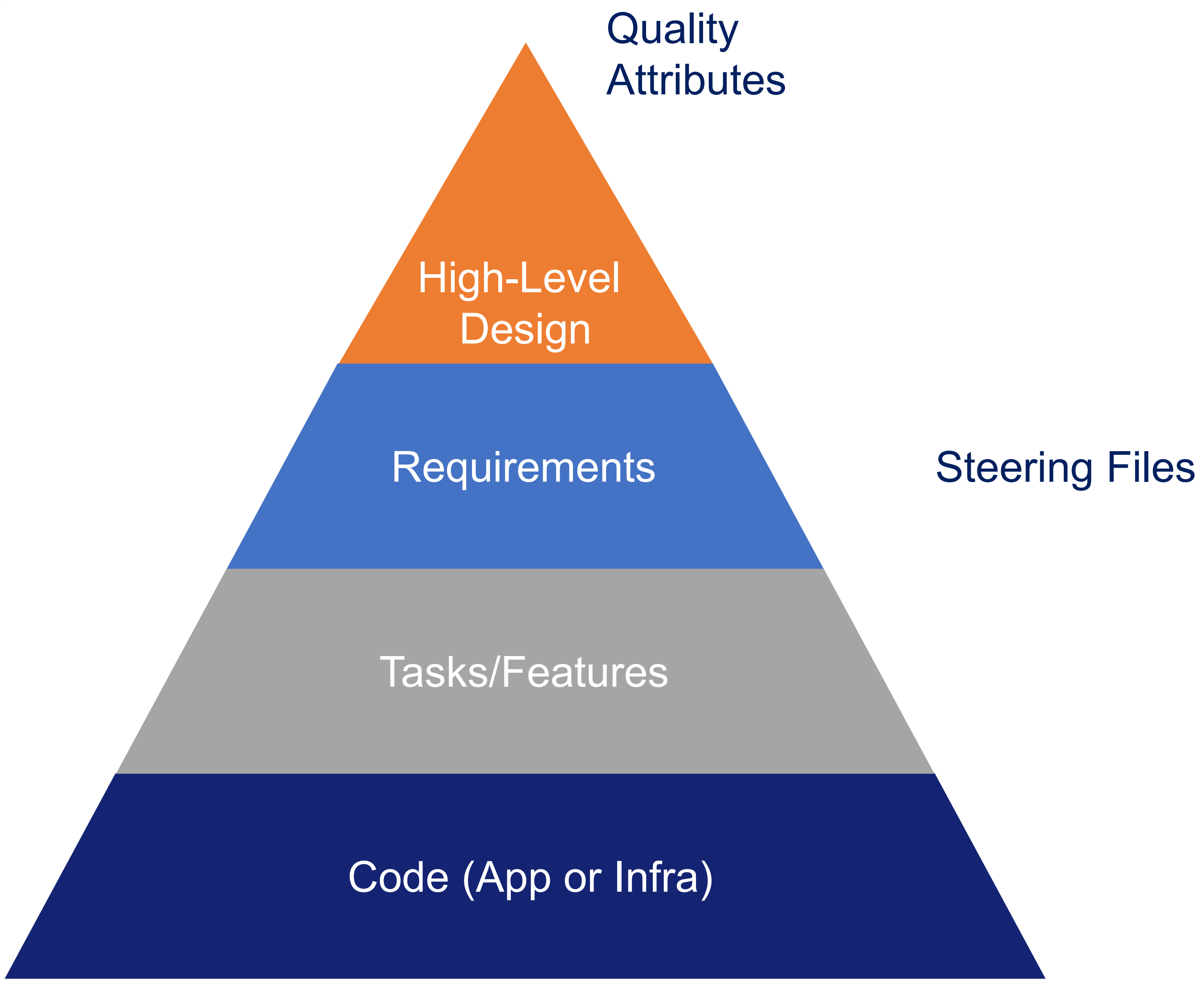
Documented and verifiable

Feature	Waterfall	Agile	Vibe Coding (Chat AI)	Spec-Drive Development (SDD)
Role of Doc	Rigid Contract	Minimal Guidance	N/A	Executable Blueprint (Living Doc)
Context Management	Contract	Human Memory	Token Window	Local Index & Spec Files
Source of Truth	Front Doc	Working Software		Living Specification
Repeatability	Components/Libraries	Manual	Hit or Miss	Deterministic Plans
Code Quality	Human PR Review	Human PR Review	Review?	Spec Validated

Needs to define...

Objective: Define an organized method to manage the migration/reorg of workloads.





Approaches for SDD

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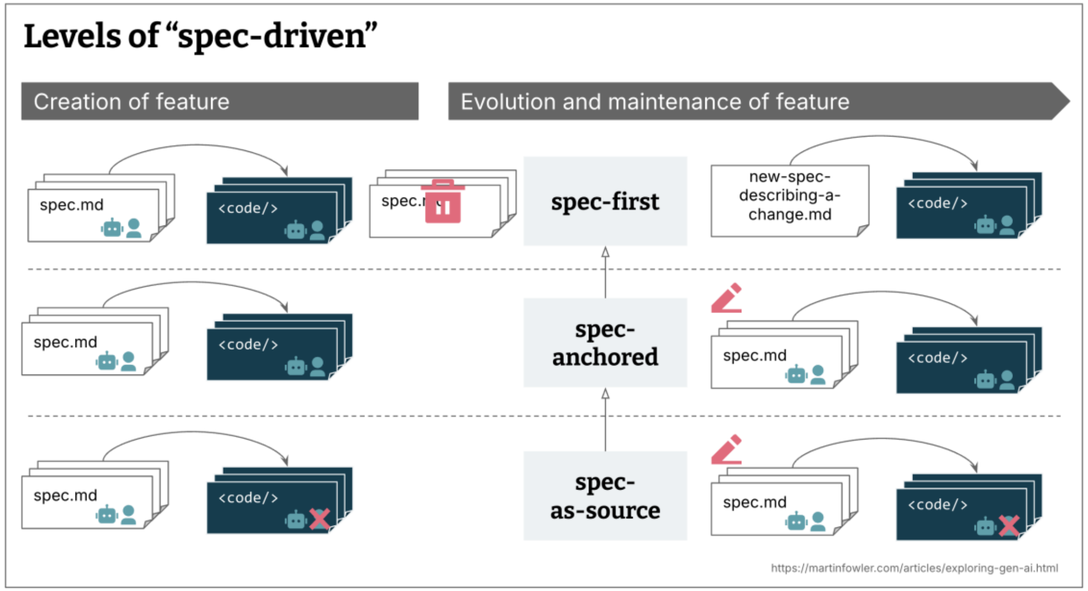
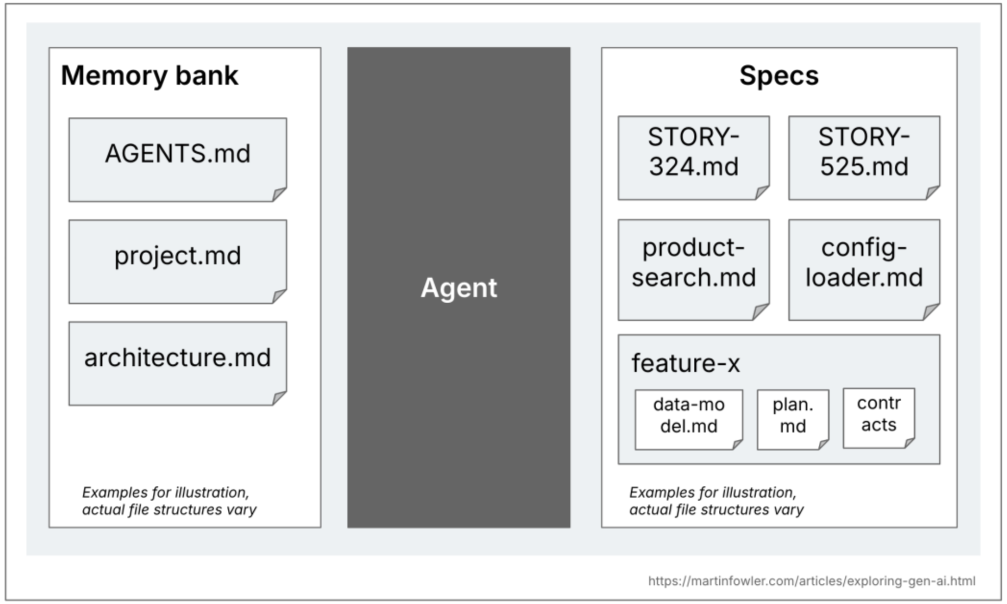
Evolution of BDD and TDD

3 approaches:
- Spec-First
- Spec-Anchored
- Spec-as-source.

Differences due to modifications on initial specifications and code.
Structured similar: Memory bank, specs, features/tasks

There are another SDD frameworks, well-mature such as GitHub SpecKit, or OpenSpec; or methodologies such as BMAD. For simplicity, I use Kiro.





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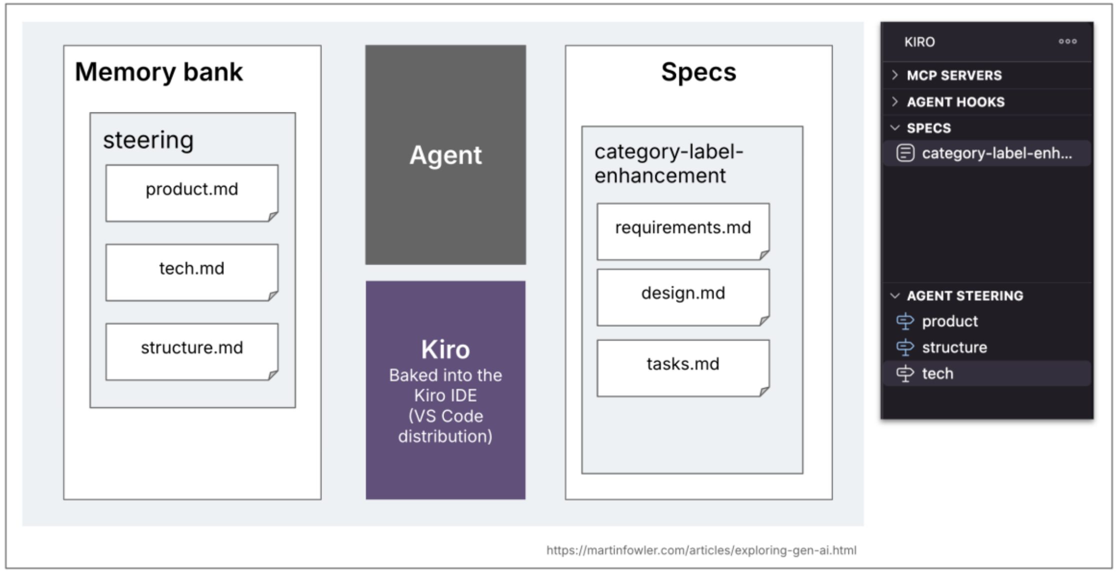
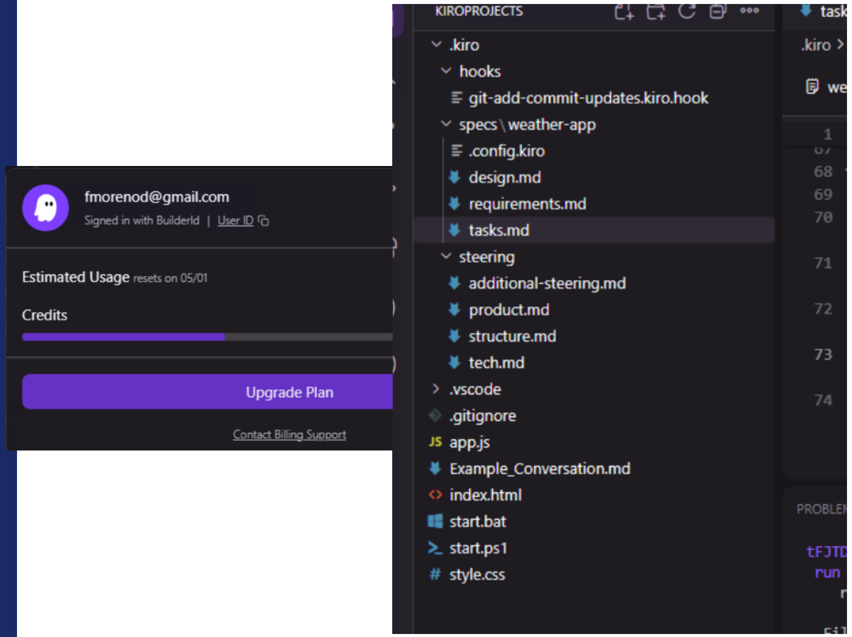
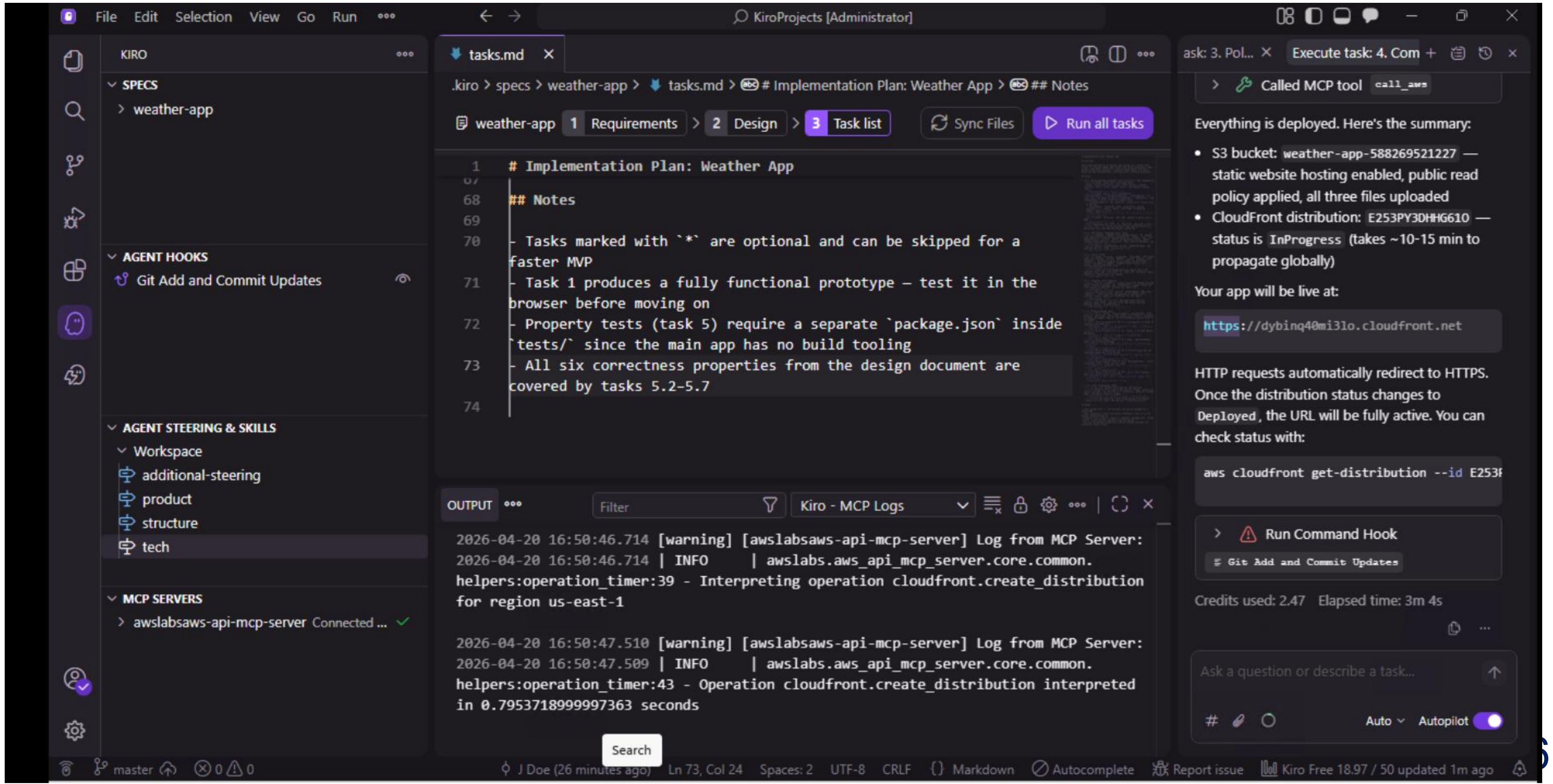
Kiro

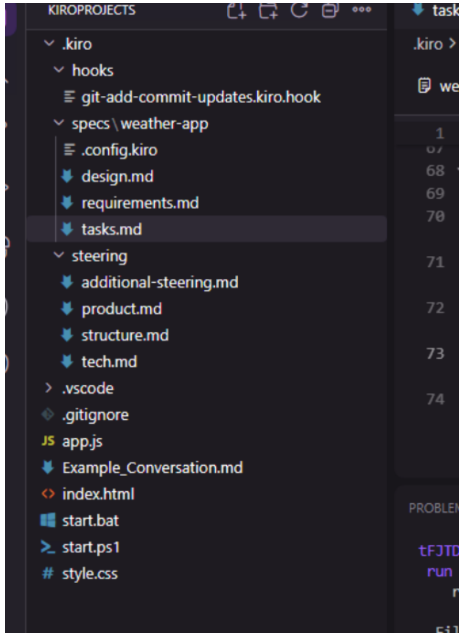
- AI IDE with focus on Spec-Driven Dev, for features but support bug fixing.
- AWS managed and charge LLM.
- AWS Support and Training
- Simple and intuitive to learn







Kiro: Feature Trinity



Initial Prompt

Requirements

- Functional Requirements. Captures user intentions and its constraints. Defined what needs to do it in EARS syntax.

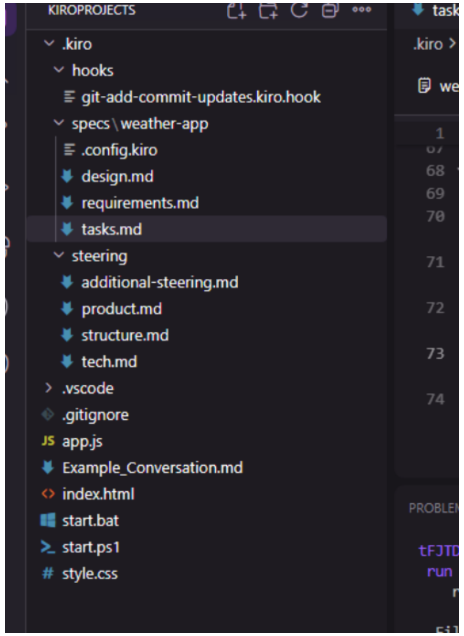
Design

- Schema abouts how to do it. State machines and relationships.

Tasks

- Executable plan. Atomic sequence plan that agent has to follow.

Kiro: Steering



Define general rules that aren't link to the feature directly. There are examples for government and tech definitions, such as:

- Tech Stack: Frontend with React and Tailwind CSS. Backend with Python 3, FastAPI. DB with PostgreSQL and Prisma as ORM
- Patterns: Using Domain-Driven Development. All code has to have a header documentation about purpose of the file and methods.
- Naming definition: Use PascalCase for components names and camelCase for file names.

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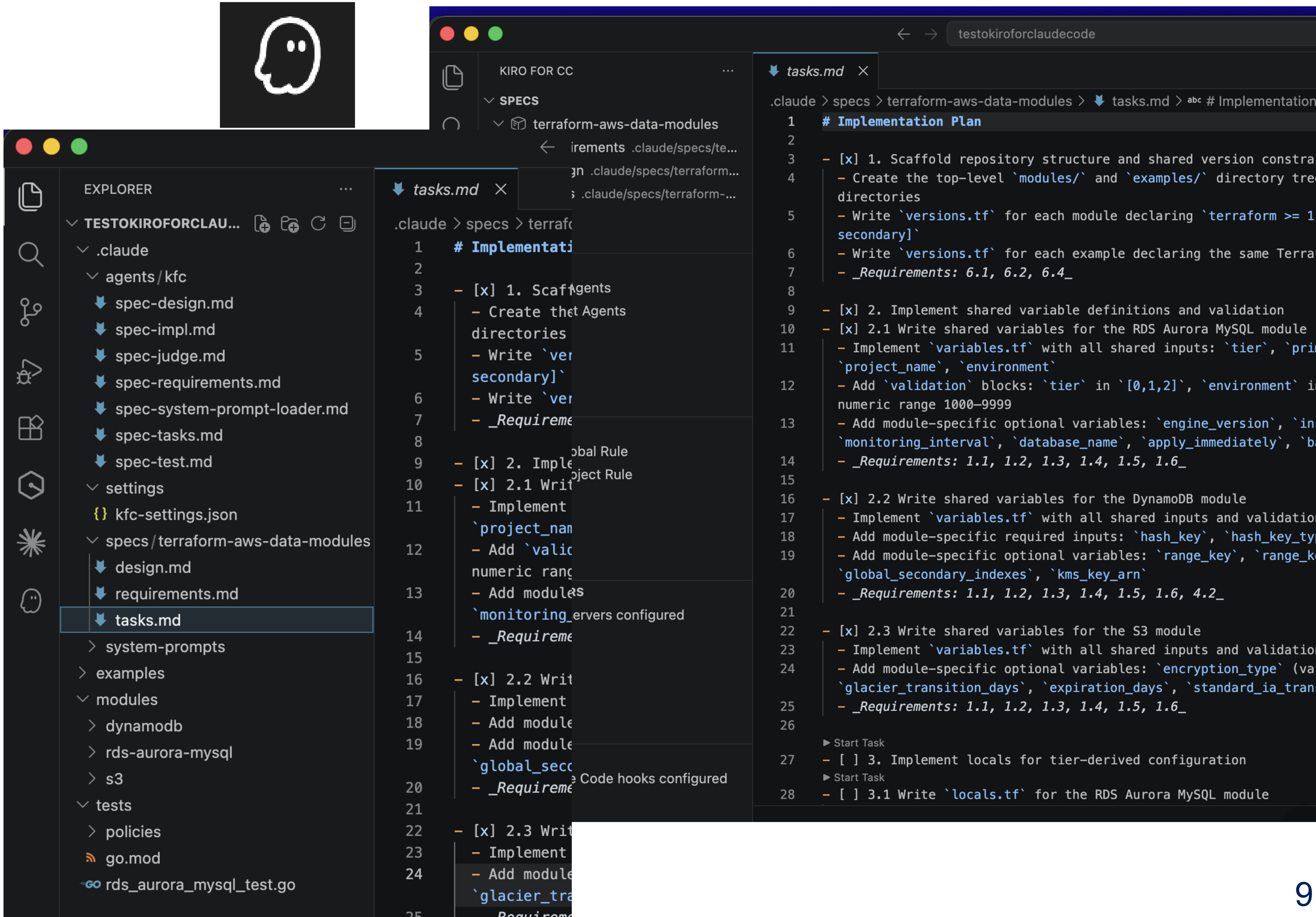
Kiro on Claude Code

- VSCode Extension
- Need a Claude Code API Keys
- Bypass Security to act as agent
- Create more under-the-hood files (.claude/agents)

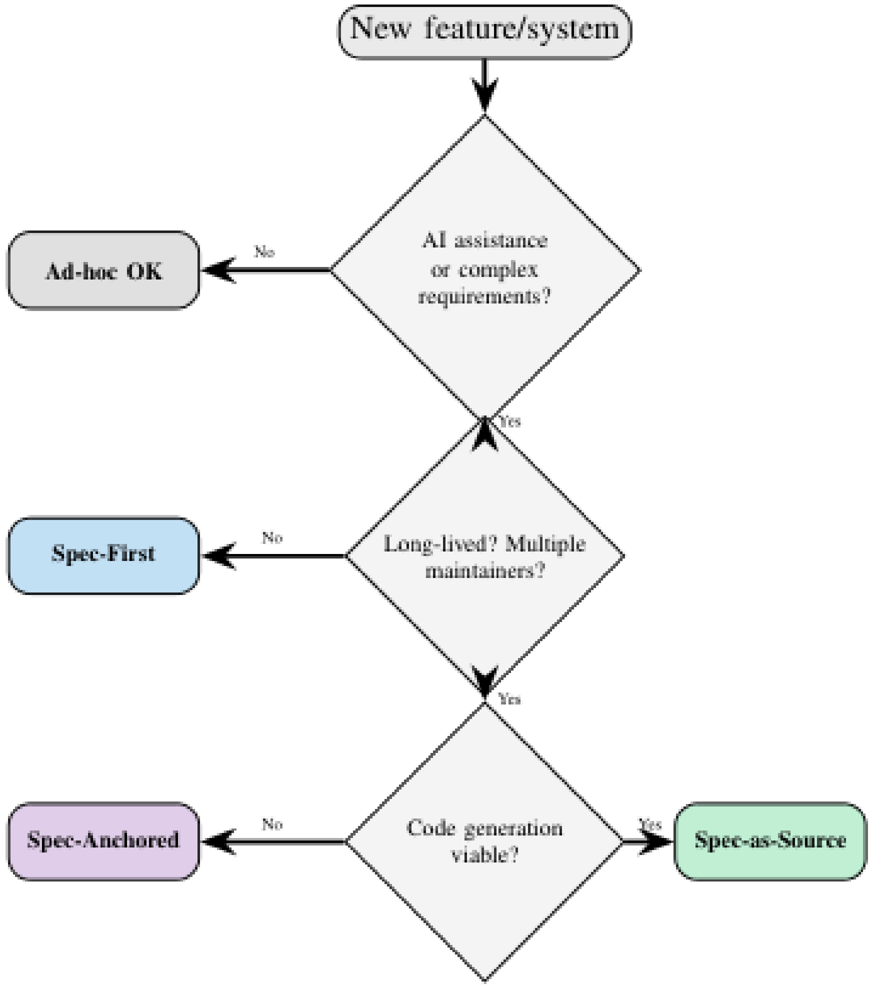


Out of Scope

My recommendations are:

- **Branching strategy:** Use gitflow for complex projects, otherwise use trunk-based strategy for simplicity.
- **Context Window:** Depends on scale (member number) and distribution of the project, it can be a problem. Try to reduce limit information on the trinity or use limited information, i.e. LLM Wiki and OpenViking
- **Deployment:** Use several type of testing on early steps -focus on integration-, to avoid rework on CI/CD pipelines.
- **Human on the loop:** Consequences of avoiding this step: Reputational damage, dismissal, legal issues, bankrupt, or layoffs.

Learned Lessons



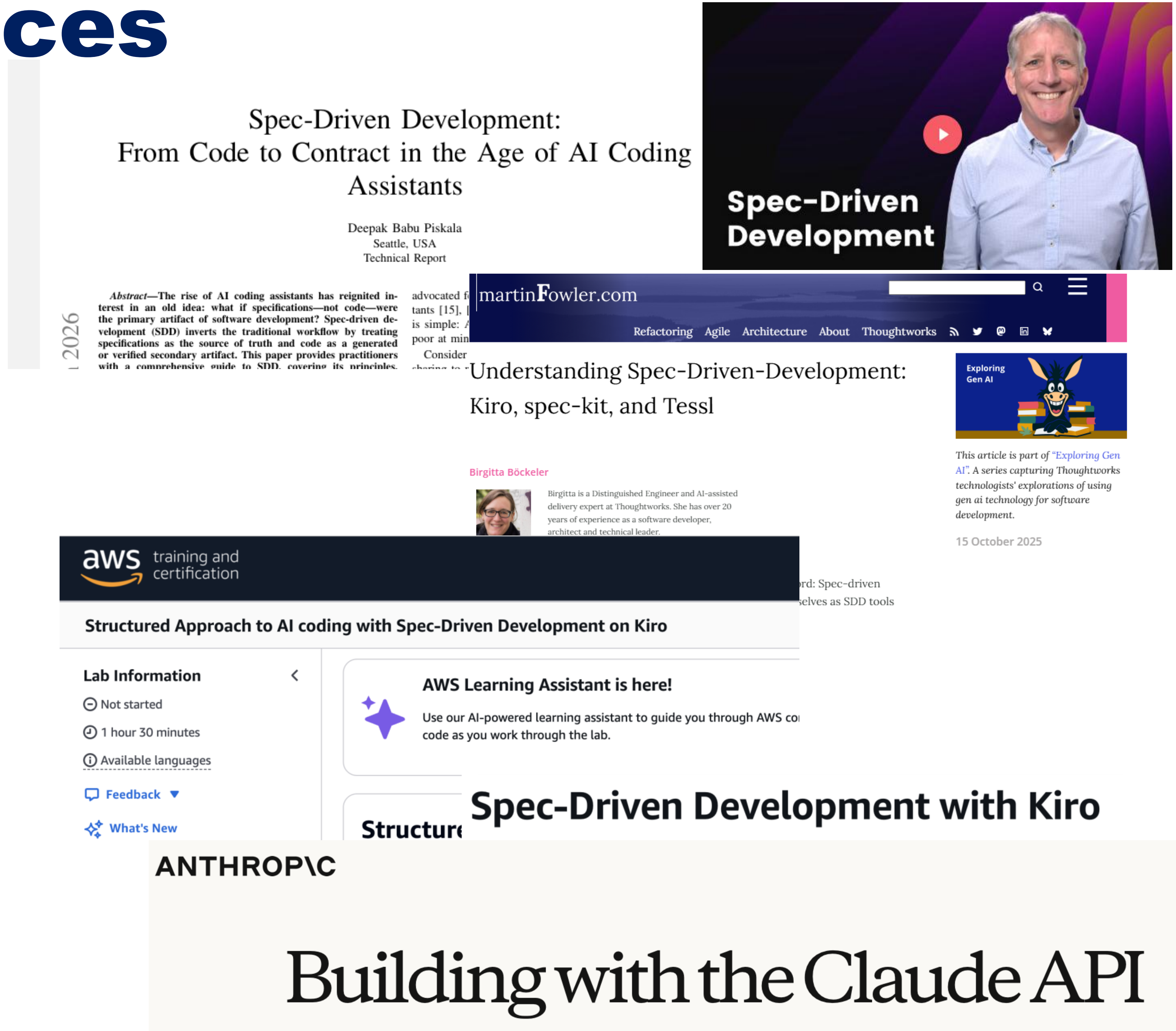
- Follow a method to scale on a team and choose the right approach based on team' maturity and maintainability.
- Need of an orchestrator (stages and transversal policies) between projects (Pipeline).
- Difference between steering and structured files (design, requirements, tasks)
- Allow optional tasks for testing (Validation tasks are longer) or using external tools.
- Modifications has to be saved on documents or on steering to have single source of truth, otherwise you can save interactions on Export Conversations.
- Cost is based on the complexity and tier.

References

About Spec-Driven Dev:
arXiV (Cornell) ,
MartinFowler and
DeepLearning

About Kiro:
SkillBuilder Courses [1, 2]

About Claude Code
Anthropic Courses [A, B]



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